

FIVE CHECKS



Before you trust AI-generated machine learning code

*"The code runs" is not the finish line.
It's the starting point.*

1

Do the parameter settings actually make sense?

Don't accept library defaults on faith.
Check that the settings match what
experienced practitioners use on problems
like yours.

2

Was there real hyperparameter tuning?

A token loop over two values isn't tuning.
Look for a meaningful search across
parameters, evaluated with cross-
validation.

3

Are the train/test splits consistent?

Every model you compare should face the same training data and the same held-out test set. Otherwise the comparison means nothing.

4

Do the metrics match your scientific question?

Accuracy and RMSE are defaults, not decisions. Pick the evaluation metric that reflects what you actually need the model to do.

5

Is there any serious interpretation?

Performance numbers are not understanding.
Ask what drives the predictions, how confident the model is, and what it means in the real world.

AI can get you from a blank file to a working prototype in minutes.

The statistical thinking is still your job.

Full post: “Five Things to Check Before You Trust AI-Generated Machine Learning Code” on the Statistical Horizons blog.